

Question number	Answer	Additional guidance	Mark
4(a)	An answer that includes two of the following points: - positive feedback results in an increase in the change (of the variable) (1) - negative feedback results in a decrease in the change (of the variable) (1)	can be phrased in many ways eg. change from normal	2

Question number	Answer	Additional guidance	Mark
4(b)	The only correct answer is D A is not the correct answer as thermoregulation is continually required. B is not the correct answer as the body continually generates heat C is not the correct answer as more heat being gained than lost means body is not in equilibrium.		1

Question number	Answer	Additional guidance	Mark
4(c)(i)	An explanation that includes the following points: - creatinine is a small molecule (1) - therefore can be forced through the {pores / basement membrane} (1) - as a result of high pressure in the glomerulus (1)	ACCEPT ultrafiltration occurs for mp1 / ignore any RMM figures allow through fenestrations	3

Question number	Answer	Additional guidance	Mark
4(c)(ii)	A calculation showing the following step: correct answer given to 3 significant figures (1)	accept $1960/15 \times 60 = 2.18 \text{ mg min}^{-1}$ $1960/15 = 131$	1

Question number	Answer	Additional guidance	Mark
4(d)(i)	An answer that includes three of the following points: - general comment about all patients after surgery - comment on difference in 2 groups - can use figures - comment using SD - comment on largest change after surgery	ACCEPT ref to error bars	3

Question number	Answer	Additional guidance	Mark
4(d)(ii)	A calculation showing the following steps: - calculation of difference surgery makes for non-diabetic and type II diabetic (1) - correct number calculated (1)	0.32 0.7 $0.7 - 0.32 = 0.38$ correct answer gains 2 marks	2